Realm comparison: AVD on Azure, Azure Local and AVD Hybrid

Thirteen rows, three deployment models. The session slide shows five of these rows (locality, lifecycle owner, identity and OS, operations, support and licensing); this handout has all thirteen.

Status: prepared 2026-10-06; cloud-only Azure Files guidance reviewed 2026-10-09 from Microsoft Learn (2609 release family for Azure Local) and the session lab; every cell was checked against the Learn pages listed under Sources. Support statements change by release, so re-check the linked documentation before you design from this table.

#	Axis	AVD on Azure	AVD on Azure Local	AVD Hybrid
1	Where compute runs	Azure	Your Azure Local cluster	Your hypervisor (Hyper-V, VMware vSphere, Nutanix AHV)
2	Lifecycle owner	Azure	Azure for the platform through Arc; you for the guest	You, with your hypervisor tools
3	Session host OS	Windows 11 Enterprise (single or multi-session); Windows Server	Windows 11 Enterprise multi-session and single-session, Windows 10 Enterprise (both), Windows Server 2025, 2022 and 2019; Azure Local 23H2 or later, registered with Azure	Windows 11 or 10 Enterprise single-session; Windows Server
4	Multi-session	Yes (Windows 11 multi-session)	Yes (Windows 11 and Windows 10 Enterprise multi-session)	No
5	Join and single sign-on	Entra join with SSO; AD DS and hybrid join also supported	The AVD portal can only join session hosts to AD DS (including hybrid join); native Entra join is supported through PowerShell and other deployment methods	Windows 11 can be Entra-only; Windows Server needs Active Directory
6	Profile storage and authentication	Azure Files: Entra Kerberos (hybrid identities in all clouds; cloud-only identities generally available in the public cloud, with client, permissions and storage-application prerequisites; external identities only for FSLogix on AVD), AD DS or Entra Domain Services; a storage account uses one identity source	FSLogix needs an SMB service, not a cluster shared volume: a file server VM or Scale-Out File Server on the cluster, or Azure Files	Any SMB path: Azure Files with Entra Kerberos, or on-premises SMB with AD DS Kerberos
7	Image tooling	Azure Image Builder to Azure Compute Gallery	Gallery or VHD, imported as an Azure Local image (there is no direct Image Builder target)	Packer with the platform builder (hyperv-iso, vsphere-iso, Nutanix plugin)

#	Axis	AVD on Azure	AVD on Azure Local	AVD Hybrid
8	Patching	Image replacement; Intune or Windows Update for Business for Windows 11. Azure Update Manager covers Windows Server, not Windows 10 or 11	Same for guests; Azure Local Lifecycle Manager for the cluster	Same for guests; hypervisor patching is yours
9	Scaling and power	Scaling plan and Start VM on Connect	Dynamic autoscale needs a session host configuration, which is not supported on Azure Local, and the Learn autoscale articles do not list Azure Local for power management; plan to manage power yourself and confirm support before relying on a scaling plan	None from AVD: Hybrid does not support power management, autoscale, Start VM on Connect or session host configuration; you scale and power-manage the VMs
10	Monitoring	AVD Insights and Azure Monitor	Same workspace, through Arc	Same workspace, through the Arc agent in the guest
11	Failed host recovery	Redeploy from the image	The cluster and the operator	The hypervisor, not AVD
12	GPU	Azure GPU VM sizes	Discrete Device Assignment (a whole GPU to one VM, no live migration) or GPU partitioning (a fraction of a GPU, one partition per VM, live migration needs OS build 26100 or later and NVIDIA vGPU 18 or later); a physical GPU is either DDA or partitioned, and the GPU configuration must be homogeneous across machines	Per hypervisor; AVD neither adds nor blocks GPU support
13	Support and licensing	Azure compute plus the user licence	Cluster, the AVD on Azure Local service fee and the user licence	Your infrastructure, the user entitlement (an eligible Windows client licence, or RDS CALs with Software Assurance for Windows Server) and the AVD Hybrid service user license; Windows multi-session is not supported on Hybrid

How to read the table

- **One workspace, three desktops:** an AVD workspace can expose desktops from all three models to the same user. Group entitlement decides which desktops the user sees; it does not decide which host the user lands on.
- **One profile pattern:** the same FSLogix configuration and the same user identity give the same container in each model, for sequential use. Roaming a profile is not replication, and replication is not recovery.
- **Shared monitoring, separate operations:** one Log Analytics workspace sees all three. Patching, power and recovery differ, as rows 8, 9 and 11 show.

Decision rules of thumb

If you need	Start with
Elasticity and scale, minimal infrastructure to run	AVD on Azure
Data, latency or regulation that keeps compute at your site	AVD on Azure Local
To reuse the hypervisor you already own, with no new hardware	AVD Hybrid

Sources

Microsoft Learn: Azure Virtual Desktop on Azure Local (overview, supported configurations, limitations, licensing and pricing); Azure Virtual Desktop for Azure Local (architecture); Azure Virtual Desktop Hybrid overview and deployment; Licensing Azure Virtual Desktop; Prepare GPUs for Azure Local, Manage GPUs via Discrete Device Assignment, Manage GPUs using partitioning; Autoscale scaling plans, glossary and scenarios; Deploy Azure Virtual Desktop (session host configuration); Enable Microsoft Entra Kerberos authentication for hybrid and cloud-only identities on Azure Files; Azure Update Manager FAQ.

Cloud-only Azure Files guidance reviewed 9 October 2026: Azure Files GA release notes, Entra Kerberos prerequisites, and supported directory ACL methods.